

PUEBLO, CO, USA

WMO: 724640

Lat: 38.289N

Lon: 104.506W

Elev: 1441

StdP: 85.17

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93058

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-17.2	-13.8	-22.0	0.6	-8.3	-19.4	0.8	-3.8	14.6	6.6	12.8	7.5	1.8	260	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	37.1	16.8	35.7	16.7	34.2	16.5	19.5	29.7	18.9	29.1	18.3	28.5	4.9	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	14.6	20.9	16.3	13.8	20.9	15.5	13.1	21.1	61.6	29.2	59.4	28.9	57.4	28.2	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.9	11.1	9.1	-23.0	39.4	3.7	1.0	-25.7	40.1	-27.9	40.7	-30.0	41.3	-32.7	42.0		
(5)			-23.4	20.8	3.6	0.6	-26.0	21.3	-28.1	21.6	-30.1	22.0	-32.7	22.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.7	-0.1	1.5	6.4	10.5	15.9	21.9	24.9	23.6	19.2	11.5	4.9	-0.3		
	DBStd	10.04	5.68	5.80	5.15	4.69	4.22	3.44	2.52	2.46	3.99	4.89	5.07	5.68		
	HDD10.0	1268	314	242	132	50	6	0	0	0	2	40	162	321		
	HDD18.3	2988	571	472	370	236	96	9	1	1	37	215	402	578		
	CDD10.0	1891	1	3	20	64	191	357	462	420	277	86	9	1		
	CDD18.3	569	0	0	0	1	21	116	204	163	62	3	0	0		
	CDH23.3	8541	1	2	32	133	584	1829	2662	1998	1074	212	14	1		
	CDH26.7	4366	0	0	2	23	225	978	1528	1053	498	57	1	0		
(14) Wind	WSAvg	3.5	3.0	3.4	3.8	4.4	4.0	3.9	3.6	3.3	3.2	3.3	3.0	3.0		
(15) Precipitation	PrecAvg	293	8	8	21	29	30	34	51	49	21	17	13	10		
	PrecMax	455	24	35	75	135	68	108	131	111	69	65	63	25		
	PrecMin	159	0	0	1	0	0	0	2	2	2	0	0	0		
	PrecStd	68	6	7	16	29	18	27	31	27	15	16	13	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.9	22.7	26.4	29.3	34.2	38.5	39.2	37.4	35.5	30.8	25.5	20.4		
		MCWB	7.4	8.1	9.9	11.2	14.7	16.1	17.4	17.2	16.2	13.0	9.7	7.4		
	2%	DB	17.4	19.4	23.8	26.8	31.4	36.5	37.5	35.7	33.6	28.3	22.4	17.0		
		MCWB	5.9	6.9	8.7	10.5	13.5	15.7	17.1	17.0	15.4	12.2	8.6	5.9		
	5%	DB	14.4	16.7	21.3	24.7	29.3	34.5	36.1	34.2	31.9	25.8	19.4	13.9		
		MCWB	4.6	5.7	7.9	9.8	13.1	15.6	17.2	17.0	15.0	11.2	7.5	4.5		
	10%	DB	11.0	13.0	18.4	22.1	26.8	32.5	34.6	32.6	29.9	22.7	16.0	10.5		
		MCWB	3.1	4.0	6.7	9.0	12.4	15.4	17.1	17.1	14.8	10.4	6.0	3.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	8.6	10.7	12.8	17.0	19.4	20.5	20.2	18.5	15.1	10.2	7.9		
		MCDB	19.4	21.2	24.3	23.8	27.9	31.1	30.1	29.6	29.6	24.3	23.3	19.4		
	2%	WB	6.2	7.2	9.4	11.6	15.5	18.2	19.7	19.4	17.4	13.5	9.0	6.2		
		MCDB	16.5	18.9	21.6	23.0	26.1	29.8	29.6	29.5	27.8	23.8	21.5	16.4		
	5%	WB	4.9	5.8	8.3	10.6	14.5	17.4	19.1	18.8	16.5	12.2	7.8	4.7		
		MCDB	13.9	16.0	20.2	22.1	24.9	28.8	29.2	28.7	27.6	23.2	18.7	13.6		
	10%	WB	3.3	4.3	7.1	9.7	13.6	16.7	18.5	18.2	15.7	11.0	6.4	3.1		
		MCDB	10.5	12.6	17.6	20.8	23.8	28.0	28.8	27.8	26.3	21.6	15.2	10.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.6	17.3	18.2	17.1	17.1	18.3	17.6	16.8	17.9	18.1	17.8	17.1		
		MCDBR	23.1	23.2	23.8	23.0	22.0	22.1	20.6	19.8	21.4	23.4	24.0	22.5		
	5% WB	MCWBR	13.3	12.6	11.6	10.0	8.3	7.4	6.1	5.7	7.5	10.3	12.4	13.1		
		MCWBR	22.0	22.5	22.3	20.5	18.6	18.2	16.3	16.2	17.7	20.1	22.6	21.6		
(40) Clear-Sky Solar Irradiance	taub		0.241	0.248	0.271	0.302	0.320	0.339	0.370	0.351	0.312	0.276	0.251	0.244		
	taud		2.614	2.614	2.543	2.459	2.466	2.429	2.410	2.456	2.527	2.601	2.603	2.594		
	Ebn at Noon		974	1004	1000	978	958	935	903	916	940	954	950	943		
	Edh at Noon		68	78	94	109	110	114	116	108	94	79	69	65		
(44) All-Sky Solar Radiation	RadAvg		2.79	3.69	4.98	6.09	6.78	7.45	7.05	6.43	5.54	4.19	3.09	2.46		
	RadStd		0.17	0.26	0.35	0.34	0.48	0.32	0.29	0.23	0.28	0.24	0.18	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+127	+98
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+75	+32

Nomenclature: See separate page